NeuroSense Tremor Classifier — Full Plan

Stage 1 — Data Loading & Subject Selection

- Load `imu_sdata.pickle` and `subject_metadata.pickle`
- Split subjects by `healthstatus_id`: PD patients (0) vs Healthy Controls (1 and 2)
- For each subject, flatten all sessions into a list of 5-second windows with metadata attached:
    - subject ID
    - health status
    - session index
    - window index within session
    - UPDRS tremor subscore if available (from `med_eval_1` / `med_eval_2`)

Stage 2 — Feature Extraction Per Window

Each window is shape `(500, 3)` — 500 samples × 3 axes at 100Hz.

Step 1 — Preprocessing

- Compute signal magnitude: `mag = sqrt(x² + y² + z²)` — collapses 3 axes into one signal, more robust to phone orientation
- Apply a high-pass filter (>1 Hz) to remove gravity component

Step 2 — Spectral Analysis

- Run FFT on the magnitude signal
- Compute Power Spectral Density (PSD)

Step 3 — Extract 3 Features

- **Tremor power** — sum of PSD in the 4–6 Hz band
- **Dominant frequency** — frequency with peak power within 3–7 Hz (slightly wider to catch edge cases)
- **Tremor regularity** — sharpness of the spectral peak, computed as ratio of peak power to mean power in the 3–7 Hz band. High ratio = narrow sharp peak = sustained rhythmic tremor

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Stage 3 — Threshold Calibration from SData

This is what makes the classifier data-driven rather than arbitrary.

- Compute tremor power for every window across **all PD subjects only**
- Build the distribution
- Set thresholds:
  - `LOW_THRESH` = 25th percentile of PD subject tremor power
  - `MID_THRESH` = 60th percentile
  - `HIGH_THRESH` = 85th percentile
- Set `REG_THRESH` = median regularity score across PD windows
- These 4 values are your classifier's only parameters — derived from data, explainable, adjustable

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Stage 4 — Per-Window Classification

Rule-based classifier using the 4 thresholds:

Condition	Label
Power < LOW or dominant freq outside 4–6 Hz	`None`
Power ≥ LOW but < MID, or regularity < REG	`Mild`
Power ≥ MID but < HIGH	`Moderate`
Power ≥ HIGH and regularity ≥ REG	`Severe`

Each window gets one label. No probabilities needed for MVP.

Stage 5 — Hourly Aggregation

- Group windows by hour using session timestamps
- **Majority vote** across all windows in that hour → one bin per hour
- Track window count per hour:
  - ≥ 6 windows → confident
  - 3–5 windows → low confidence flag ⚠
  - < 3 windows → mark as no data, don't show a label

- Hours with no sessions at all → grey / no data in the view

Stage 6 — ON/OFF Derivation

Simple mapping on top of tremor bins — no separate classifier needed:

Tremor Bin	Motor State
`None`	ON
`Mild`	ON
`Moderate`	OFF
`Severe`	OFF

This gives you the ON/OFF timeline as a byproduct of the tremor classifier, not a separate model.

Stage 7 — Validation Against UPDRS

Before building the view, sanity check the classifier:

- For SData subjects with UPDRS tremor subscores, check that:
  - High UPDRS tremor score → majority of their windows labeled Moderate/Severe
  - Low UPDRS score → majority labeled None/Mild
- This is not a formal accuracy metric — it's a face validity check for the hackathon
- Report the agreement rate as your "clinical grounding" claim in the pitch

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Stage 8 — Output Structure

What gets passed to the clinician report view:

Per hour:
- hour timestamp

- tremor bin: None / Mild / Moderate / Severe
- motor state: ON / OFF
- window count + confidence flag
- dominant frequency (for tooltip detail)

Per day summary:
- % hours in each bin
- % ON-time
- peak tremor hour
- data coverage (hours with data / 24)

Stage 9 — Clinician Report View

7-Day Timeline Row (one row per day, 24 columns)

Color	Meaning
Green	ON (None/Mild tremor)
Amber	Moderate
Red	Severe / OFF
Grey	No data

Daily Summary Card

- ON-time %
- Tremor severity breakdown as a small bar (% hours in each bin)
- Peak tremor period (e.g. "Most severe: 8–10am")
- Data coverage caveat if < 50% of hours have data

LLM Narrative Input

- Pass the daily summaries as structured JSON
- The LLM converts them into the 3-paragraph clinical narrative
- Example input signal: "Monday: 60% ON, peak Moderate tremor 8–10am, 6 hours of data"

What This Is Not Claiming

Important for pitch honesty:

- This is not a diagnostic tool — PD diagnosis is pre-established
- Thresholds are population-derived from SData, not clinically validated cutoffs
- Phase 2 replaces the rule-based thresholds with a trained classifier using UPDRS as ground truth
- Hourly bins assume session data is representative of that hour — not guaranteed with SData's call-based recordings